6.3 Logarithms and Logarithmic Functions

Name _____ Period ____ Date _____

Algebra II 6.3 (30-minute plan) · Student sheet

I can…

- rewrite logarithmic and exponential equations.
- evaluate logarithms, with and without a calculator.
- find inverses and graph logarithmic functions.

Vocabulary · fill in the missing word

Logarithm — $\log_b y = x$ _____ $b^x = y$ ($b > 0$, $b \neq 1$): the exponent you put on b to get y

Common logarithm — $\log x$ = _____

Natural logarithm — $\ln x$ = _____

Logarithmic function — $f(x) = \log_b x$, the inverse of $g(x) = b^x$; domain $x > 0$, _____ $x = 0$

Part 1 · Notes and examples (fill in each note, then work the example)

Definition of a logarithm: $\log_b y = x$ means _____. A logarithm is an _____.

Example 1

a. Rewrite $\log_3 81 = 4$ in exponential form. b. Rewrite $6^{-1} = \frac{1}{6}$ in logarithmic form.

Special logarithms: $\log_b 1$ = _____ $\log_b b$ = _____ log x means log base _____ ln x means log base _____

Example 2

Evaluate with a calculator. Round to three decimal places.

a. log 8 b. ln 0.3

Inverse properties: Inverse properties: $\log_b b^x$ = _____ and b^($\log_b$ x) = _____

Logarithmic functions: $f(x) = \log_b x$ is the inverse of _____. Its domain is _____ and its asymptote is _____.

Example 3

Find the inverse of each function.

a. $y = 6^x$ b. $y = \ln(x + 3)$

Part 2 · Partner practice

Partner practice

Evaluate $\log_2 32$ and $\log_7 \left(\frac{1}{49}\right)$.